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GUIDANCE AND EVENT PREDICTION FOR WELL DRILLING OPERATIONS

Abstract

A method for use with a subterranean well drilling operation can include training a predictive model with historical well data to predict a probability of a historical wellbore event occurring, then inputting to the trained predictive model parameters of a target well to be drilled, and the predictive model predicting a probability of a wellbore event occurring in the target well. A system for use with a subterranean well drilling operation can include a predictive model trained to predict at least one historical wellbore event, based on historical well data. The predictive model can be configured to predict a probability of a wellbore event occurring in a target well. The predictive model can be configured to provide guidance for drilling the target well.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application claims the benefit of the filing date of U.S. provisional application No. 63/558,353 filed on 27 Feb. 2024. The entire disclosure of the prior application is incorporated herein by this reference in its entirety.

BACKGROUND

[0002] This disclosure relates generally to equipment utilized and operations performed in conjunction with a subterranean well and, in an example described below, more particularly provides for drilling guidance and prediction of drilling events for a well drilling operation.

[0003] Undesirable drilling events (such as, stuck pipe, wellbore collapse, lost circulation, gas influx, etc.) can result in costly non-productive time and added expense. Unfortunately, it can be very difficult to predict when undesirable drilling events will occur. This also makes it very difficult for an operator to take appropriate steps to prevent an undesirable drilling event, or to be prepared with to take appropriate steps to mitigate an undesirable drilling event if it should occur.

[0004] Therefore, it will be readily appreciated that improvements are continually needed in the art of drilling a wellbore. Such improvements can be useful in preventing undesirable drilling events from occurring, and/or in mitigating the undesirable effects of drilling events that have occurred. Techniques described herein for use in drilling a wellbore may be performed prior to drilling the wellbore, and/or in real time while the wellbore is being drilled.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. 1 is a representative partially cross-sectional view of an example of a well system and associated method which can embody principles of this disclosure.

[0006] FIG. 2 is a representative flowchart for an example method of determining drilling mud rheology that may be used with the FIG. 1 system and method.

[0007] FIG. 3 is a representative flowchart for an example method of training a predictive model, based on historical cuttings data and wellbore events.

[0008] FIG. 4 is a representative flowchart for an example method of using a trained predictive model to predict wellbore events and provide guidance for a target well.

[0009] FIG. 5 is a representative flowchart for an example method of training a predictive model, based on historical well data, in conjunction with geomechanics principles, to predict wellbore events and provide guidance for a target well.

[0010] FIG. 6 is a representative flowchart for an example method of using a trained predictive model, in conjunction with geomechanics principles, to predict wellbore events in real time and provide guidance in real time, while a target well is being drilled.

[0011] FIG. 7 is a representative schematic view of an example system for use with a drilling operation, in which a predictive model is trained using offset well data.

[0012] FIG. 8 is a representative schematic view of the example system of FIG. 7, in which the trained predictive model is used to provide guidance and predict wellbore events for a target well.

[0013] FIG. 9 is a representative sectioned view of an example stratigraphy with certain stresses indicated.

[0014] FIG. 10 is a representative schematic view of an example mechanical earth model.

[0015] FIGS. 11-13 are example main logs for three offset wells.

[0016] FIGS. **14-16** are representative rock property profiles estimated by the model for the three offset wells.

[0017] FIGS. **17A & B** are representative pore pressure calibrations for two of the offset wells.

[0018] FIG. **18** is a representative flowchart for a process of determining maximum horizontal stress.

[0019] FIG. **19** is a representative depiction of a range of mud weights and their corresponding effects on wellbore stability.

[0020] FIGS. **20A & B** is a representative mud weight window predicted for an example offset well.

[0021] FIGS. **21A-C** is a further example of a representative mud weight window predicted for an offset well.

[0022] FIG. **22** is an additional representative mud weight window predicted for an offset well.

[0023] FIGS. **23A-C** is a representative log depicting synthetic wellbore stability results generated for an example target well.

DETAILED DESCRIPTION

[0024] Representatively illustrated in FIG. **1** is a system **10** for use with a subterranean well and an associated method which can embody principles of this disclosure. However, it should be clearly understood that the system **10** and method are merely one example of an application of the principles of this disclosure in practice, and a wide variety of other examples are possible. Therefore, the scope of this disclosure is not limited at all to the details of the system **10** and method described herein and/or depicted in the drawings.

[0025] In the FIG. **1** example, a wellbore **12** is being drilled into or through an earth formation **14**. A jointed or continuous drill string **16**, including drill pipe **18**, has a drill bit **20** connected at a distal end thereof. The drill bit **20** may be rotated from the earth's surface (for example, using a rotary table or top drive of a drilling rig), and/or the drill bit may be rotated by a drill motor **22** of a bottom hole assembly **24** at the distal end of the drill string **16**.

[0026] The bottom hole assembly **24** can include any of a wide variety of different types of equipment used in a drilling operation. For example, the bottom hole assembly **24** can include a measurement while drilling (MWD) and/or logging while drilling (LWD) tool **26**, a centralizer, a caliper, a stabilizer, a reamer, a rotary steering tool, etc. The scope of this disclosure is not limited to any particular tools, elements or combinations thereof used in the bottom hole assembly **24**.

[0027] As viewed in FIG. **1**, the wellbore **12** is being drilled vertically (i.e., with no inclination relative to vertical). In other examples, all or portions of the wellbore **12** could be drilled with a non-zero inclination relative to vertical. The scope of this disclosure is not limited to any particular trajectory or inclination of the wellbore **12**.

[0028] As depicted in FIG. **1**, the formation **14** includes bedding planes **28** having a dip angle α relative to horizontal. The wellbore **12** in this example is not orthogonal to the bedding planes **28**, but in other examples the wellbore **12** could be orthogonal to the bedding planes, and/or the bedding planes could have a dip angle α of zero. The scope of this disclosure is not limited to any particular relative orientation of the wellbore **12** and bedding planes **28**.

[0029] A drilling fluid or drilling “mud” **30** is circulated through the drill string **16** during the drilling operation. The drilling mud **30** exits via nozzles in the drill bit **20** and returns to the surface via an annulus **32** formed between the drill string **16** and the wellbore **12**.

[0030] The drilling mud **30** performs many functions in the drilling operation. Due to its density, the drilling mud **30** produces hydrostatic pressure in the wellbore **12**, which prevents (or at least mitigates) an influx of fluids into the wellbore from the formation **14**. The drilling mud **30** cools and lubricates the drill bit **20**. The drilling mud **30** typically includes additives, such as barite, that forms a “mud cake” on an inner wall of the wellbore **12**, to thereby stabilize the wellbore and at least mitigate fluid transfer between the wellbore and the formation **14**. The drilling mud **30** also suspends drill cuttings **34** in the annulus **32** and transports the cuttings to the surface. However, the

scope of this disclosure is not limited to use of the drilling mud **30** to perform any particular function, or combination of functions, in the drilling operation.

[0031] It is beneficial to be able to determine certain properties of the drilling mud **30** in a drilling operation. In particular, rheological properties (such as dynamic viscosity μ) can be useful, for example, to estimate the cuttings removal capacity of the drilling mud **30**, and to aid in calculation of equivalent circulating density of the drilling mud and, thus, bottom hole pressure in the wellbore **12** during the drilling operation.

[0032] Referring additionally now to FIG. **2**, a representative flowchart for an example method **36** of determining drilling mud rheology is representatively illustrated. For convenience, the method **36** is described below as it may be used with the FIG. **1** system **10** and method. However, the FIG. **2** method **36** may be used with other systems or methods in keeping with the principles of this disclosure.

[0033] In an initial step **38** of the method **36**, laboratory rheology data is input to a computational device, such as a computer, capable of mathematical analysis. In this example, the laboratory rheology data includes viscometer measurements (such as, measurements output by a Fann 35 viscometer) for a variety of different mud densities at a variety of different temperatures and/or different pressures. That is, for each mud density, viscometer measurements are recorded at multiple different temperatures and/or pressures. For each mud density, and at each temperature and/or pressure, viscosity measurements are taken at different rotational speeds of the viscometer.

[0034] The mud types (oil based mud and/or water based mud), mud densities and temperatures/pressures are preferably chosen to correspond to a range of those expected to be used or experienced in a typical drilling operation. For example, mud densities of about 9 to about 16 pounds per gallon, and temperatures of about 75 to about 140 degrees F., may be used in the above viscosity measurement step. Of course, other mud densities and temperatures may be used in keeping with the scope of this disclosure.

[0035] In step **40**, the recorded viscosity measurements can then be used to produce a correlation model for each mud density (also known to those skilled in the art as mud “weight”) of a given mud type. For example, a straight line fit can be determined for the viscosity measurements at each mud density. Such a straight line fit can have the form of $\mu = x(T) + y$, in which μ is dynamic viscosity, T is temperature, and x and y are constants determined for each mud density. Similarly, linear correlations between dynamic viscosity and pressure can be built following the same approach. For varying temperatures and pressures, a 3D plane fit can be used to make viscosity predictions. Such plane can have the form of $\mu = x(T) + y(p) + z$, in which μ is dynamic viscosity, T is temperature, p is pressure, and x , y and z are constants determined for each mud density. More complex (non-linear) correlations can also be built as $\mu = f(T, p)$. In this manner, for a given mud density, a dynamic viscosity can be determined at any expected temperature and pressure. This approach can apply to rheology properties other than viscosity.

[0036] The correlation model can be used to aid in planning future well drilling operations, and can be incorporated into a well drilling control system for use in real time while a well is being drilled. For example, the mud density can change while a wellbore is being drilled, due to various changing conditions (such as, a change in overburden gradient, an under-pressurized zone, etc.). It is, therefore, beneficial to be able to determine changes in mud rheology in real time.

[0037] In step **42**, a given mud weight (mud density) is selected. The selected mud weight can be one of the mud densities for which laboratory rheology data was recorded for step **38**. The selected mud density will correspond to one of the correlation models produced in step **40**.

[0038] In step **44**, a temperature and/or a pressure is entered. The entered temperature and/or pressure can then be used by the correlation model to determine the mud rheology in step **46**. If the correlation model is a straight line fit to the laboratory rheology data as described above, the dynamic viscosity μ will be estimated as $x(T) + y$, where T is the entered temperature, and x and y are the constants previously determined for the correlation model. Similarly, linear correlations

between dynamic viscosity and pressure can be built following the same approach. For varying temperatures and pressures, a 3D plane fit can be used to make viscosity predictions. Such plane can have the form of $\mu = x(T) + y(p) + z$, in which μ is dynamic viscosity, T is temperature, p is pressure, and x , y and z are constants determined for each mud density. More complex (non-linear) correlations can also be built as $\mu = f(T, p)$. In this manner, for a given mud density, a dynamic viscosity can be determined at any expected temperature and pressure.

[0039] Referring additionally now to FIG. 3, a flowchart for an example method **48** of training a predictive model is representatively illustrated. In the FIG. 3 example, the predictive model is trained using historical cuttings data and wellbore events.

[0040] As mentioned above, the drilling mud **30** performs various functions in a well drilling operation. The present inventors propose that cuttings data (e.g., sizes, shapes and orientations of cuttings transported to the surface by the drilling mud while drilling) can be used to train a predictive model to predict whether wellbore events (such as, wellbore instability, wellbore collapse, stuck pipe, etc.) will occur in a drilling operation.

[0041] The predictive model can be any type of machine learning or artificial intelligence. Neural networks, genetic algorithms, etc., may be used in the predictive model.

[0042] In step **50**, historical cuttings data is input to the predictive model. The historical cuttings data may be recorded in end of well reports generated for previous well drilling operations. The previously drilled wells may preferably be wells in a same field, such as offset wells, or the previously drilled wells may not be in the same field.

[0043] The historical cuttings data may be recorded by manually measuring drill cuttings **34**, for example, at a shale shaker used with a drilling rig. As another alternative, one or more cameras may be used to detect cuttings quantity, shape, size, orientation, etc. automatically during a drilling operation.

[0044] The end of well reports generated for the previous well drilling operations will also include any wellbore events (such as, wellbore instability, wellbore collapse, stuck pipe, etc.) that occurred during the drilling operations. The end of well reports will also typically include operational data, such as, drilling depth, trajectory, formation lithology (including rock type and dip angle α of bedding planes **28**), pore pressure, mud density, overburden gradient, field, year, temperature, etc.

[0045] In step **52**, the predictive model is trained to predict the historical wellbore events, based on the historical cuttings data input in step **50**. The predictive model is sufficiently trained, for example, when it can estimate the probability that a given wellbore event will occur during a drilling operation, based on the historical cuttings data input in step **50**, and the estimated probability is acceptably matched to a wellbore event that occurred in the previous drilling operation.

[0046] In step **54**, well parameters for a target well to be drilled are input to the trained predictive model. The target well parameters may be the same type as, or similar to, the types of operational data described above as being included in the historical well data (e.g., trajectory, depth, rate of penetration, lithology, expected pore pressure, planned mud density, expected overburden gradient, field, year, expected temperature, etc.).

[0047] In step **56**, the trained predictive model will predict the possibilities that wellbore events will occur while drilling the target well. For example, the predictive model could predict the probability of a wellbore collapse, gas influx or stuck pipe if the target well is drilled using the parameters input in step **54**. If the probabilities of undesirable wellbore events predicted by the predictive model are acceptably low, then the operator may decide to proceed with drilling the target well using the parameters input in step **54**.

[0048] In step **58**, the predictive model can be used to provide guidance for drilling the target well, for example, if in step **56** the predictive model predicts an unacceptably high probability that an undesirable wellbore event will occur. For example, if the predictive model predicts a high probability that a wellbore collapse will occur due to insufficient drilling mud density, then the mud

density input in step **54** can be increased (such as, incrementally), until the predictive model no longer predicts an unacceptably high probability that the wellbore collapse will occur. As another example, if the predictive model predicts a high probability that stuck pipe will occur due to an inappropriate orientation of the wellbore **12** relative to the bedding planes **28**, then the trajectory of the wellbore can be changed, until the predictive model no longer predicts an unacceptably high probability that the stuck pipe will occur.

[0049] Referring additionally now to FIG. **4**, a flowchart for an example method **60** of using a trained predictive model to predict wellbore events and provide guidance for a target well is representatively illustrated. The predictive model used in the FIG. **4** method **60** is preferably the predictive model that was trained and used to provide guidance for drilling the target well in the FIG. **3** method **48**.

[0050] However, it should be noted here that the predictive model can continue to be trained before, during and after the target well is drilled. At any point additional well drilling operational data is available, that data can be input to the predictive model for further training of the predictive model. After the current target well is drilled, the trained predictive model can be used to evaluate and provide guidance for drilling a further target well, and can continue to be trained during the drilling of that further target well.

[0051] In step **62** of the FIG. **4** method **60**, cuttings data (e.g., quantity, size, shape, bedding plane orientation, etc.) is input to the trained predictive model in real time while the target well is being drilled. Preferably, additional operational data (such as, depth, mud density, caliper, rate of penetration, temperature, pore pressure, etc.) is also input to the trained predictive model as the target well is being drilled.

[0052] In step **64**, the predictive model predicts in real time the probabilities of undesirable wellbore events occurring during the drilling of the target well, based on the inputs of step **62** and the prior training of the predictive model. It is expected that these probabilities will change over time due, for example, to changing conditions as the target well is being drilled. Therefore, preferably the operator monitors the probabilities predicted by the predictive model as the target well is being drilled.

[0053] In step **66**, the predictive model can provide guidance for actions to be taken while the target well is being drilled. For example, if an unacceptably high probability of an undesirable wellbore event occurring is predicted in step **64**, then in step **66** the predictive model can be used to determine what action should be taken to mitigate that possibility. In some examples, the predictive model may cause an alarm to be activated if an unacceptably high probability of an undesirable wellbore event is predicted, and may indicate to the operator what action should be taken to prevent or at least mitigate the occurrence of the undesirable wellbore event.

[0054] For example, wellbore instability and, therefore, a relatively high probability of wellbore collapse or stuck pipe, can be indicated by cavings production at the shale shaker. Cavings are drill cuttings that typically (but not always) have a relatively large size compared to normal cuttings. Types of cavings include splintered, angular and tabular cavings.

[0055] Splintered cavings can occur in shales as a result of drilling under balance (insufficient mud density, with tensile failure). Splintered cavings can also occur due to drilling rapidly (high rate of penetration) through low permeability shales, and failure due to dilatancy. If splintered cavings are present, then in step **66** the predictive model could provide guidance to increase the mud density, and/or reduce the rate of penetration.

[0056] Angular cavings can occur due to high stress concentration at the wellbore wall exceeding the rock strength, resulting in shear failure. If angular cavings are present, then in step **66** the predictive model could provide guidance to gradually increase the mud density. Of course, the mud density would need to be maintained within a safe range to avoid inadvertently fracturing the formation **14**, undesirable fluid loss or influx, etc.

[0057] Tabular cavings can occur due to drilling across the formation bedding planes **28** at an

unfavorable angle of attack (e.g., nearly parallel to the dip inclination and perpendicular to the dip direction), or due to drilling across a fractured formation. If tabular cavings are present, then in step **66**, the predictive model could provide guidance to change the drilling trajectory.

[0058] The production of cavings, their type and size could be part of the cuttings data input to the predictive model during its training, and in real time while the target well is being drilled (in steps **50** and **62**). In the FIG. **4** method **60**, the predictive model can predict in step **64** the probability of a wellbore event (such as, wellbore collapse, wellbore instability, stuck pipe, etc.), based in part on the cuttings data, including whether and what type of cavings are being produced. In step **66**, the predictive model can provide guidance on how any predicted undesirable wellbore event can be prevented or mitigated, or could have been prevented or mitigated in the past (for example, as part of a post-job analysis).

[0059] Referring additionally now to FIG. **5**, a flowchart for an example method **68** of training a predictive model, based on historical well data is representatively illustrated. In the FIG. **5** method **68**, the training of the predictive model is performed in conjunction with geomechanics principles, to predict wellbore events and provide guidance for a target well to be subsequently drilled.

[0060] The method **68** is similar in many respects to the FIG. **3** method **48**, in that historical data is used to train the predictive model. In the FIG. **5** method **68** the historical well data input to the predictive model (step **70**) includes all relevant data that is available for previously drilled wells. This historical well data can include depths, lithology (including rock type and dip orientation), pore pressure, mud density, overburden gradient, field, year drilled, trajectory, caliper, cuttings data, temperature, rate of penetration, vibration data, torque, any wellbore events that occurred, etc. The scope of this disclosure is not limited to any particular data or combination thereof used to train the predictive model in the method **68**.

[0061] In step **72**, the predictive model is trained to predict the historical wellbore events. This may be an iterative process, in which adjustments are made to the predictive model with each iteration, so that the predictive model becomes progressively more accurate in predicting when the historical wellbore events occurred.

[0062] When the predictive model is suitably trained, then in step **74** known or planned parameters for the target well are input to the predictive model. The target well parameters may be the same type as, or similar to, the types of operational data described above as being included in the historical well data (e.g., trajectory, depth, rate of penetration, lithology, expected pore pressure, planned mud density, expected overburden gradient, field, year, expected temperature, etc.).

[0063] In step **76**, the trained predictive model will predict the possibilities that wellbore events will occur while drilling the target well. For example, the predictive model could predict the probability of a wellbore collapse, gas influx or stuck pipe if the target well is drilled using the parameters input in step **74**. If the probabilities of undesirable wellbore events predicted by the predictive model are acceptably low, then the operator may decide to proceed with drilling the target well using the parameters input in step **74**.

[0064] Note that, in the FIG. **5** method **68**, geomechanics principles **78** are used with the predictive model in its predictions of wellbore events in step **76**. The geomechanics principles **78** can also be used with the predictive model to predict pore pressure and a safe mud density window for the target well.

[0065] In step **80**, the predictive model can be used to provide guidance for drilling the target well, for example, if in step **76** the predictive model predicts an unacceptably high probability that an undesirable wellbore event will occur. For example, if the predictive model predicts a high probability that a wellbore collapse will occur due to insufficient drilling mud density, then the mud density input in step **74** can be increased (such as, incrementally), until the predictive model no longer predicts an unacceptably high probability that the wellbore collapse will occur. As another example, if the predictive model predicts a high probability that stuck pipe will occur due to an inappropriate orientation of the wellbore **12** relative to the bedding planes **28**, then the trajectory of

the wellbore can be changed, until the predictive model no longer predicts an unacceptably high probability that the stuck pipe will occur. The geomechanics principles **78** can be used with the predictive model in step **80** to provide appropriate guidance for the drilling of the target well. [0066] The predictive model can predict a safe mud density window, pore pressure and other parameters for the target well, based on the inputs of step **74**. Such predictions output by the predictive model can be based also on the geomechanics principles **78**.

[0067] Referring additionally now to FIG. **6**, a flowchart for an example method **82** of using a trained predictive model is representatively illustrated. In the FIG. **6** method **82**, the trained predictive model of the FIG. **5** method **68** is used in conjunction with geomechanics principles to predict wellbore events in real time and to provide guidance in real time, while a target well is being drilled. Prediction of wellbore events using the method **82** can also be used in planning the target well drilling operation and in post-job analysis.

[0068] However, it should be noted here that the predictive model can continue to be trained before, during and after the target well is drilled. At any point additional well drilling operational data is available, that data can be input to the predictive model for further training of the predictive model. After the current target well is drilled, the trained predictive model can be used to evaluate and provide guidance for drilling a further target well, and can continue to be trained during the drilling of that further target well.

[0069] In step **84** of the FIG. **6** method **82**, well data is input to the trained predictive model in real time while the target well is being drilled. The well data can include depth, mud density, caliper, rate of penetration, temperature, pore pressure, vibration, torque, trajectory, well location (for example, including proximity to offset wells), etc. The scope of this disclosure is not limited to any particular well data or combination thereof input in real time to the predictive model.

[0070] In step **86**, the predictive model predicts in real time the probabilities of undesirable wellbore events occurring during the drilling of the target well, based on the inputs of step **84**, the geomechanics principles **78** and the prior training of the predictive model. It is expected that these probabilities will change over time due, for example, to changing conditions as the target well is being drilled. Therefore, preferably the operator monitors the probabilities predicted by the predictive model as the target well is being drilled.

[0071] In step **88**, the predictive model can provide guidance for actions to be taken while the target well is being drilled. The geomechanics principles are also used by the predictive model in step **88** for providing the guidance. For example, if an unacceptably high probability of an undesirable wellbore event occurring is predicted in step **86**, then in step **88** the predictive model can be used to determine what action should be taken to mitigate that possibility. In some examples, the predictive model may cause an alarm to be activated if an unacceptably high probability of an undesirable wellbore event is predicted, and may indicate to the operator what action should be taken to prevent or at least mitigate the occurrence of the undesirable wellbore event.

[0072] Referring additionally now to FIG. **7**, a schematic view of an example system **90** for use with a drilling operation is representatively illustrated. The predictive model **90** may be used with any of the methods **36**, **48**, **60**, **68** described herein, or with other methods. In the FIG. **7** system **90**, a predictive model **92** is trained using offset well data.

[0073] In the FIG. **7** example, historical well data **94** from the drilling of the offset wells is input to the predictive model **92** in the training process. The historical well data **94** can include any available type of well data, such as well data included in end of well reports, LWD data, well location, etc. The scope of this disclosure is not limited to any particular types of well data or combination thereof used to train the predictive model **92**.

[0074] The predictive model **92** may include any type of model capable of being trained to predict parameters (such as, the probability of a wellbore event occurring, pore pressure, a safe mud density window, etc.) for the target well. The predictive model **92** may be constructed by

combining all or some of the models previously described. The predictive model **92** can include artificial intelligence, neural networks, genetic algorithms and/or any type of predictive engine. Geomechanics principles **78** can be used with the predictive model **92** in the training process.

[0075] Referring additionally now to FIG. **8**, a schematic view of the example system **90** of FIG. **7**, in which the trained predictive model **92** is used to provide guidance and predict wellbore events for a target well, is representatively illustrated. In the FIG. **8** example, real time well data **96** is input to the trained predictive model **92** as the target well is being drilled.

[0076] The real time well data **96** can include any of the types of well data described above, such as, depth, lithology, pore pressure, mud density, overburden gradient, field, year, trajectory, caliper, cuttings data, temperature, downhole pressure, well location, vibration, torque, fluid influx, fluid loss, etc. The scope of this disclosure is not limited to any particular types of real time well data or combination thereof input to the trained predictive model **92** as the target well is being drilled.

[0077] As depicted in FIG. **8**, the trained predictive model **92** can provide guidance, predict a probability of a wellbore event occurring and predict parameters (such as, pore pressure, horizontal stresses in the formation **14**, and a safe mud density window) in real time while the target well is being drilled. The predictive model **92** can produce an alert or alarm if the predicted probability of an undesirable wellbore event occurring is unacceptably high. The guidance provided by the predictive model **92** can include changes in mud density, trajectory, rate of penetration, etc., in order to prevent or mitigate undesirable wellbore events (such as, wellbore instability, wellbore collapse, stuck pipe, etc.). Geomechanics principles **78** can be used with the predictive model **92** in providing the real time guidance and predictions.

[0078] It may now be fully appreciated that the above disclosure provides significant advancements to the well drilling art. In examples described above, the predictive model **92** can be used, along with geomechanics principles, to predict probabilities of wellbore events occurring in a target well to be drilled. The predictive model **92** can also be used to predict certain parameters (such as, a safe mud density window, pore pressure, horizontal stresses, etc.) for the target well, and to provide guidance both before the target well is drilled and in real time during the target well drilling operation.

[0079] The above disclosure provides to the art a method for use with a subterranean well drilling operation. In one example, the method can comprise: training a predictive model **92** with historical well data **94** to predict a probability of a historical wellbore event occurring; then inputting to the trained predictive model **92** parameters of a target well to be drilled; and the predictive model **92** predicting a probability of a wellbore event occurring in the target well.

[0080] The historical well data **94** may comprise data recorded for at least one offset well.

[0081] The predictive model **92** may comprise artificial intelligence. In addition, or in the alternative, neural networks, genetic algorithms, machine learning or other types of artificial intelligence may be used.

[0082] The predicting step may include providing geomechanics principles to the predictive model **92**.

[0083] The method can include the predictive model **92** predicting a safe mud density window for the target well. The method can include the predictive model **92** predicting a pore pressure for the target well.

[0084] The historical well data **94** may comprise historical drill cuttings data.

[0085] The method may include inputting to the predictive model **92** well data for the target well in real time as the target well is being drilled.

[0086] The method may include the predictive model **92** providing guidance for drilling the target well as the target well is being drilled.

[0087] The method may include producing a correlation model for a relationship between mud density, temperature and mud rheology.

[0088] The above disclosure also provides to the art a system **90** for use with a subterranean well

drilling operation. In one example, the system **90** can comprise a predictive model **92** trained to predict at least one historical wellbore event, based on historical well data **94**. The predictive model **92** is configured to predict a probability of a wellbore event occurring in a target well. The predictive model **92** is configured to provide guidance for drilling the target well.

[0089] The historical well data **94** may include drill cuttings data.

[0090] The predictive model **94** may comprise artificial intelligence.

[0091] The guidance may include a safe mud density window.

[0092] The predictive model **92** may be configured to predict the probability of the wellbore event occurring in the target well, based in part on geomechanics principles **78** provided to the predictive model.

[0093] The predictive model **92** may be configured to provide the guidance for drilling the target well, based in part on geomechanics principles **78** provided to the predictive model.

[0094] The historical well data may be recorded for at least one offset well.

[0095] The predictive model **92** may comprise a correlation model for a relationship between mud density, temperature and mud rheology.

[0096] The historical well event may comprise at least one of the group consisting of wellbore collapse, wellbore instability and stuck pipe.

[0097] The predictive model **92** may be configured to predict horizontal stresses in an earth formation **14** of the target well.

[0098] Although various examples have been described above, with each example having certain features, it should be understood that it is not necessary for a particular feature of one example to be used exclusively with that example. Instead, any of the features described above and/or depicted in the drawings can be combined with any of the examples, in addition to or in substitution for any of the other features of those examples. One example's features are not mutually exclusive to another example's features. Instead, the scope of this disclosure encompasses any combination of any of the features.

[0099] Although each example described above includes a certain combination of features, it should be understood that it is not necessary for all features of an example to be used. Instead, any of the features described above can be used, without any other particular feature or features also being used.

[0100] The terms “including,” “includes,” “comprising,” “comprises,” and similar terms are used in a non-limiting sense in this specification. For example, if a system, method, apparatus, device, etc., is described as “including” a certain feature or element, the system, method, apparatus, device, etc., can include that feature or element, and can also include other features or elements. Similarly, the term “comprises” is considered to mean “comprises, but is not limited to.”

[0101] Of course, a person skilled in the art would, upon a careful consideration of the above description of representative embodiments of the disclosure, readily appreciate that many modifications, additions, substitutions, deletions, and other changes may be made to the specific embodiments, and such changes are contemplated by the principles of this disclosure. For example, structures disclosed as being separately formed can, in other examples, be integrally formed and vice versa. Accordingly, the foregoing detailed description is to be clearly understood as being given by way of illustration and example only, the spirit and scope of the invention being limited solely by the appended claims and their equivalents.

Geomechanics Principles

[0102] Following and in FIGS. **9-23C** is a description of basic geomechanics principles, and examples of how geomechanics principles can be incorporated into a mechanical earth model (MEM). An MEM is a numerical representation of the state of stresses, pore pressure and rock mechanical properties for a specific stratigraphic section. It contains all relevant geomechanical information for the analysis of wellbore instabilities, sand production, hydraulic fracturing and production-induced reservoir deformations.

[0103] FIG. **9** depicts an example of a stratigraphic section, with certain stresses indicated. In FIG. **9**, $S_{\text{sub.v}}$ is vertical stress, $S_{\text{sub.Hmax}}$ is maximum horizontal stress, $S_{\text{sub.hmin}}$ is minimum horizontal stress, and P_p is pore pressure. In addition, $S_{\text{sub.HmaxAzi}}$ is stress direction, and UCS is rock strength.

[0104] FIG. **10** depicts a mechanical earth model and its components (derived from Plumb et al., 2000). FIG. **10** depicts a flow chart listing the steps in an example method to generate and calibrate an MEM. Specific details of the generation of multiple example MEMs are described more fully below. The techniques described herein have provided internally consistent example MEMs that integrated and honored all of the available data.

[0105] After data collection and analysis, the MEM construction starts by defining the lithostratigraphy followed by calculations of rock elastic properties and rock strength. After that, geo-pressure and stresses are calculated. Stresses and failure criterion are estimated with the poro-elastic approach.

[0106] The MEM enables a reproduction of wellbore failures observed by the means of a caliper and/or borehole image in order to estimate the state of stress responsible for these failures. The example MEMs described herein were constructed for wells designated WELL_X1, WELL_X2 and WELL_X3, allowing for geomechanical analyses of wellbore stability for these wells in prediction of drilling each planned well.

Model Assumptions & Uncertainties

[0107] Effect of discontinuities (like bedding failure, highly laminated layers) etc. [0108] Assumption of 1-Dimensional isotropic and homogeneous model. [0109] Lack of core test calibration data to estimate accurate mechanical properties across formation of interest. The rock properties were estimated using correlations. [0110] No frac-data and formation pressure test to calibrate minimum horizontal stress magnitude. The fracture gradient was calibrated using a formation integrity test (FIT). [0111] 1D model has limitations to predictions in compartmentalization areas. So, extrapolation of the data has inherited uncertainties. [0112] The observed incidents and the caliper log were used to calibrate the shear failure gradient (breakout line). [0113] If the distance between offset wells and target well is relatively large, this will result in high uncertainty due high variation in rock properties, anisotropy and compartmentalization. [0114] The offset wells were drilled around 10 years ago so the pore pressure prediction for the target well will have high uncertainty due to reservoir depletion.

[0115] FIGS. **11-13** depict examples of main logs used in the analysis for three offset wells.

[0116] Rock mechanical properties are an important and essential parameter for predicting wellbore failure, constraining in situ stress magnitudes and predicting mud weight for safe drilling operations. Rock strength refers to the ability of the rock to withstand the stress environment. Rock behavior under stress is not only dependent on the strength, but also to the rock elastic constants. These parameters, all together, define how a rock behaves under different loading conditions and when it fails.

[0117] The following rock properties are preferably used to populate a geomechanical model:

[0118] Unconfined Compressive Strength (UCS_COM).Math. [0119] Friction Angle (FANG_LAL).Math. [0120] Static Young's modulus (YME_STA_LA).Math. [0121] Poisson's ratio (PR_STA)

[0122] The formation rock properties were calculated using well-known empirical equations relating rock properties to petrophysical logs.

Elastic Moduli Calibration

[0123] Dipole sonic tool measurements (which provide compressional and shear wave travel times) along with bulk density can be used to establish the dynamic elastic rock properties which are related to rock strength and measured static moduli.

[0124] The relevant dynamic elastic moduli relating to these examples are determined from the following equations:

$$[00001] E_{dyn} = \frac{4 - 3 \frac{\Delta t_s^2}{\Delta t_c^2}}{\Delta t_s^2 \cdot \text{Math.} (1 - \frac{\Delta t_s^2}{\Delta t_c^2})} \text{Young's Modulus dynamic (psi)} \times 1.34 \times 10^{10}$$

$$= \frac{2 - \frac{\Delta t_s^2}{\Delta t_c^2}}{(1 - \frac{\Delta t_s^2}{\Delta t_c^2})} \text{Poisson Ratio dynamic}$$

[0125] Where: ρ : Bulk Density, Δt_c : Compressional Slowness, Δt_s : Shear Slowness

[0126] Dynamic elastic properties consider the material is perfectly elastic and then provide higher values of rock stiffness, actual rock fabric contains flaws and plane of weakness that decreases the stiffness of the rock and then a conversion from static properties is required. In this case, empirical equations for dynamic to static conversion were applied and then these were slightly modified to match the results from the triaxial tests.

Rock Strength

[0127] In addition to the rock elastic properties discussed above, rock strength parameters are also required to complete the characterization of the mechanical response and load bearing capacity of the rock. The tensile strength is the limit to the capacity of the rock to resist stretching and it is estimated to be 10% of the UCS as demonstrated by tensile tests in laboratory. There are several numbers of empirical equations presented by different researchers and companies for specific regions or rock types. Due to limitations associated with the application of these equations, it is important to calibrate them with laboratory data on local samples.

[0128] In this study, the correlations used to estimate rock strength are shown in the following table:

TABLE-US-00001 LITHOLOGY EQUATION AUTHOR(S) REMARKS LIMESTONE $143.8 \exp(-6.95\Phi)$ Chang et al, Limestones with 2006 low to moderate porosity and relatively high strength SHALE $1.35 \cdot (304.8/DTC)_{sup.2.6}$ Chang et al, Globally 2006 applicable

Unconfined Compressive Strength Correlations Used

[0129] The limitation to the above correlations is that UCS values are obtained for each respective lithology only and thus, they do not give a continuous log profile. Composite, continuous UCS_COM logs for each lithology were obtained by calculating the results of the individual correlations and then combining the results based on data availability, then the composite UCS_COM logs for each lithology was merged into one.

[0130] FIGS. 14-16 show the rock properties profiles estimated by the model for all three offset wells.

Overburden Stress

[0131] The overburden stress at given depth point is the weight of overlying earth material and can be calculated by integrating a density log for the above formations. Typically, formation density is obtained from wireline logs and can be supplemented with values obtained from core measurements. Intervals of missing or poor log quality are extrapolated using an exponential curve:

$$[00002] \rho_b = \rho_{mud} + A_0 \times (TVD - AG - WD)$$

[0132] Where $\rho_{sub.b}$, $A_{sub.0}$ and a are three fitting parameters, $\rho_{sub.mud}$ is mud density, TVD is true vertical depth, WD is water depth and AG is air gap.

[0133] For these examples, the extrapolation method was used to generate a synthetic density which is extrapolated up to mud line using the above exponential equation with three control points.

[0134] The overburden stress is determined by integrating the densities of all rock layers above a given depth of interest using the equation below.

$$[00003] \sigma_v = \int_0^z \rho_b \cdot \text{Math.} (z) \cdot \text{Math.} g \cdot \text{Math.} dz$$

[0135] Where $\sigma_{sub.v}$ is overburden stress, ρ is density, z is depth, and g is gravity

Pore Pressure

[0136] Pore pressure can be determined using a variety of techniques including direct measurement and inferences from mud weight and logging data. These measurements are typically done in permeable formations when fluid samples are taken using a wireline formation testing tool. Sonic

and resistivity logs can be used to identify pore pressure trends which can be used to estimate the pore pressure in shale formations. The estimated pore pressure needs to be calibrated by pore pressure data. FIGS. 17A & B depict example pore pressure calibrations for the example wells WELL_X2 and WELL_X1.

[0137] For WELL_X2, the DST points 2752 psi and 2777 psi were used to calibrate the pressure profile at 2000 m and 2100 m respectively. For WELL_X1, MDT Pressure 5148.8 psi at 3236 m was used. Due to Missing Pp estimates of planned well formations, the usual practice of taking the same trend of Pp GRADIENT from the nearest well and calibrate it was not used. Since WELL_X3 is the nearest well and Formation G formation is at a similar depth and one side of the fault is the same for WELL_X3 and the target well TARGET_WELL_XX, the WELL_X3 Pp profile was used to calculate the gradient. Using this gradient, the Pressure Profile for TARGET_WELL_XX is calculated.

Stresses Orientation

[0138] The minimum and maximum horizontal stresses are principal stresses and thus their directions are perpendicular to each other. The direction of the horizontal stresses can be estimated from analysis of borehole breakouts on image logs and caliper data or earthquake focal mechanisms like fault directions. Breakout is characterized by symmetrical hole enlargements that are aligned along the minimum horizontal stress in vertical wells, while drilling induced tensile fractures (DITF), if developed, are in maximum horizontal stress direction.

[0139] In the example target area for the wells, the horizontal stresses are compressional. Maximum horizontal stresses are in a NW-SE direction, and the stress regime of the target area is predominantly strike slip and thrust regime.

Horizontal Stress Magnitude

[0140] The drilling process in subsurface rock formations results in the removal of an already stressed in situ rock material. As a result, the stress originally carried by the removed material is redistributed in the surrounding rock around the wellbore. This stress redistribution causes a stress concentration, which is higher than the original in situ earth stresses.

[0141] If the stress concentration exceeds the strength of the surrounding rock, a wellbore wall failure occurs. At a given depth below ground surface and for a known wellbore azimuth inclination, the stress concentration is a function of the original in situ stresses. Knowing the accurate and precise magnitude of in situ stresses is, therefore, a basic and fundamental step to make a reasonable wellbore stability prediction. The poro-elastic horizontal strain model was used to model the magnitudes of the minimum and maximum horizontal stress in these examples.

[00004]

$$\begin{aligned} \sigma_h &= b \cdot P_p + \frac{1}{1-\alpha} \cdot (\sigma_v - b \cdot P_p) + \frac{E}{1-\alpha} \cdot \Delta \epsilon_H + \frac{E \cdot \alpha}{1-\alpha^2} \cdot \Delta \epsilon_h \\ \sigma_H &= b \cdot P_p + \frac{1}{1-\alpha} \cdot (\sigma_v - b \cdot P_p) + \frac{E \cdot \alpha}{1-\alpha^2} \cdot \Delta \epsilon_H + \frac{E}{1-\alpha} \cdot \Delta \epsilon_h \end{aligned}$$

[0142] Where σ_{h} is minimum horizontal stress, σ_H is maximum horizontal stress, σ_v is overburden stress, α is Biot elastic constant, P_p is pore pressure, and ϵ_H and ϵ_h are strains in minimum horizontal stress and maximum horizontal stress direction respectively. Information of wellbore breakouts, and rock strength from the offset wells WELL_X3, WELL_X2 & WELL_X1 was used to constrain the magnitude of the maximum horizontal stress. The minimum horizontal stress (fracture gradient) was calibrated using available LOT and FIT data.

[0143] As direct measurement of the maximum horizontal stress is not possible, it is inferred with reasonable accuracy through modelling like that for minimum horizontal stress but using additional constraints from wellbore failure. Shear failure of the rock under compressive stress is mainly ruled by the anisotropy of horizontal stresses.

[0144] The magnitude of the maximum horizontal stress must be selected in order to simulate the shear failure (breakouts) detected by image logs and observed in the field cavings while drilling. To determine maximum horizontal stress through modelling, the example process depicted in a FIG.

18 flowchart was used, and repeated until failure simulations were consistent with downhole records. This process is called “back analysis.”

[0145] Wellbore stability analysis can be conducted from MEM results. The stress concentrations around the borehole are calculated with the MEM as input data, and the principal stresses around the borehole can then be compared to the rock failure criteria to determine whether the borehole wall is failed or not.

[0146] One output of the wellbore stability analysis is a safe mud weight window. General speaking, there are four critical values in defining the safe mud weight window: [0147] Pore pressure. If mud weight is less than the pore pressure, wellbore washout and even well blowout can be expected. The consequences can be very severe. [0148] Minimum mud weight for preventing breakout (Shear Failure Gradient). If the mud weight is less than this value, borehole breakout will initiate, because the mud support to the borehole wall is not enough. [0149] Minimum horizontal stress. If natural fractures or any other conductive fissures exist around the wellbore, the mud pressure that is higher than the minimum horizontal stress will reopen the natural fractures/fissures, and cause drilling fluid losses, and possibly borehole collapse if the spacing of the natural fractures/fissures is small. [0150] Formation breakdown pressure. When the mud pressure is higher than the formation breakdown pressure, hydraulic fractures will be induced on the borehole wall. This can cause significant mud losses, particularly if the mud pressure is also higher than the minimum horizontal stress.

[0151] The ideal mud weight should be higher than the pore pressure and the minimum mud weight for preventing breakout, and should be lower than the minimum horizontal stress and the formation breakdown pressure.

[0152] FIG. **19** depicts a mud weight range defined by the breakout limit (lower bound) and the fracture propagation limit (upper bound). The breakout limit is the minimum mud weight that could be used without exceeding the maximum allowable breakout size, also defined by the pore pressure if the formation is appropriately strong. The fracture propagation limit or fracture gradient is equal to the magnitude of the closure stress (minimum principal stress, $S_{sub.hmin}$).

1. WELL_X1 Mud Weight Window

[0153] If the mud weight (MW) is less than the shear failure gradient (breakout), rocks will fail. In contrast, if the MW is greater than the breakout gradient, rock will not fail.

[0154] FIGS. **20A & B** depict the mud weight window of WELL_X1. Shear failure has been calibrated with a caliper log. WELL_X1 had a gas inflow at 3529 m, as can be seen from where the MW was less than the pore pressure. Hence the MW was increased from 1.2 sg to 1.46 sg to bring the well under control.

2. WELL_X3 Mud Weight Window

[0155] FIGS. **21A-C** depict the mud weight window of WELL_X3. The shear failure line is calibrated with a caliper. The FIT of 1.47 sg and 1.22 sg for Formation F formation at 1619.18 m and Formation H at 2501 m respectively was calibrated with the fracture gradient line. This is a predominantly strike slip regime. Caliper data was missing for Formation E, sand and Formation G formations.

3. WELL_X2 Mud Weight Window

[0156] FIG. **22** depicts the mud weight window of WELL_X2. The WELL_X2 well started flowing from 2501-2794 m. From the 1D MEM it is observed that MW was less than the Pore Pressure, finally mud weight was increased from 8.8 to 8.9 ppg. The shear failure matches with the caliper log. The stresses are predominantly strike slip.

[0157] The table below lists some important information for the planned well obtained from the drilling program.

TABLE-US-00002 Well Data COMMENTS Well type Vertical Target Formation Formation G, Formation I Risk Breakouts, Tight spots, Losses and stuck pipe

[0158] FIGS. **23A-C** depict the synthetic wellbore stability results generated for

TARGET_WELL_XX (planned well) by merging rock mechanical properties from WELL_X3, WELL_X2 and WELL_X1. In addition to this, rock properties and stress field profiles estimation are shown from Formation A to Formation I. From TARGET_WELL_XX, 1D MEM it is observed that Formation E, Formation H & Formation I formation is expected to be one of the weak areas during drilling, and good drilling practices must be taken into account with proper hole cleaning procedures.

[0159] In the table below are mud weight recommendations for the TARGET_WELL_XX target well:

TABLE-US-00003	MW	MW	ppg	ppg	Formation (min)	(max)	Risks	Actions
FORMATION A	8.8	9.2	Losses	Good LCM	FORMATION B	8.8	9.2	Losses
FORMATION C	8.8	9.2	Losses	Good LCM	FORMATION D	8.8	9.2	Losses
FORMATION E	9.2	9.5	Stuck pipe;	Good drilling practices, control ROP, ECD & Avoid break outs	FORMATION F	10.3	10.6	Losses; break
FORMATION G	9.2	9.7	Kick, Breakouts	Increase MW	FORMATION H	9.2	9.5	Breakouts
FORMATION I	9.5	9.7	Breakouts	Review Cavings Morphology; Increase ECD	FORMATION I	9.5	9.7	Breakouts

[0160] In the table below are in situ stress gradients and pore pressure gradients predicted for the TARGET_WELL_XX target well:

TABLE-US-00004	SHMax	Shmin	Sv	Pp	FG	Formation (psi/ft)	(psi/ft)	(psi/ft)	(psi/ft)	(ppg)
FORMATION A	1.6	0.7	1.06	0.39	13.75	FORMATION B	1.51	0.7	1.1	0.39
FORMATION C	1.9	0.78	1.13	0.39	14.5	FORMATION D	1.43	0.61	1.14	0.44
FORMATION E	1.41	0.79	1.09	0.47	14.83	FORMATION F	1.41	0.75	1.13	0.45
FORMATION G	1.41	0.738	1.1	0.46	13.4	FORMATION H	1.33	0.65	1.1	0.46
FORMATION I	1.32	0.66	1.06	0.44	11.8					

[0161] Recommendations for the TARGET_WELL_XX (planned well) are as follows: [0162] Lost Circulation: [0163] i. Mud additives loss control material (LCM) or polymeric LCM. [0164] Stuck Pipe: [0165] i. Carefully monitor cavings, cuttings and surface drilling parameters for indications of hole collapse. Increase MW if hole conditions dictate, avoid back-reaming if it is possible. [0166] ii. Good hole cleaning practices and controlled drilling are very important. [0167] iii. Avoid swabbing or surging the formation. [0168] iv. Treat the drilling fluid to improve inhibition, transport cuttings and reduce fluctuation in downhole pressures. [0169] Minimum Horizontal Stress (S.sub.hmin): Further frac tests/LOT/XLOT after each casing shoe is recommended to understand the variation in S.sub.hmin. [0170] Maximum Horizontal Stress (S.sub.HMax): No image logs available for constraining S.sub.Hmax magnitude. It is recommended to have image logs to better constrain S.sub.Hmax. [0171] Due to the complexity of the area, uncertainty and lack of offset well data close to TARGET_WELL_XX (planned well), it is highly recommended to do Real Time Monitoring in this well to validate the results of the geomechanics model. (Sonic log and Pressure data is updated continuously in real time) [0172] Using Managed Pressure Drilling is highly recommended in areas with losses, kicks and low mud weight window

Claims

1. A method for use with a subterranean well drilling operation, the method comprising: training a predictive model with historical well data to predict a probability of a historical wellbore event occurring; then inputting to the trained predictive model parameters of a target well to be drilled; and the predictive model predicting a probability of a wellbore event occurring in the target well.
2. The method of claim 1, in which the historical well data comprises data recorded for at least one offset well.
3. The method of claim 1, in which the predictive model comprises artificial intelligence.
4. The method of claim 1, in which the predicting comprises providing geomechanics principles to the predictive model.

5. The method of claim 1, further comprising the predictive model predicting a safe mud density window for the target well.
 6. The method of claim 1, further comprising the predictive model predicting a pore pressure for the target well.
 7. The method of claim 1, in which the historical well data comprises historical drill cuttings data.
 8. The method of claim 1, further comprising inputting to the predictive model well data for the target well in real time as the target well is being drilled.
 9. The method of claim 8, further comprising the predictive model providing guidance for drilling the target well as the target well is being drilled.
 10. The method of claim 1, further comprising producing a correlation model for a relationship between mud density, temperature and mud rheology.
 11. A system for use with a subterranean well drilling operation, the system comprising: a predictive model trained to predict at least one historical wellbore event, based on historical well data, in which the predictive model is configured to predict a probability of a wellbore event occurring in a target well, and in which the predictive model is configured to provide guidance for drilling the target well.
 12. The system of claim 11, in which the historical well data includes drill cuttings data.
 13. The system of claim 11, in which the predictive model comprises artificial intelligence.
 14. The system of claim 11, in which the guidance comprises a safe mud density window.
 15. The system of claim 11, in which the predictive model is further configured to predict the probability of the wellbore event occurring in the target well, based in part on geomechanics principles provided to the predictive model.
 16. The system of claim 11, in which the predictive model is further configured to provide the guidance for drilling the target well, based in part on geomechanics principles provided to the predictive model.
 17. The system of claim 11, in which the historical well data is recorded for at least one offset well.
 18. The system of claim 11, in which the predictive model comprises a correlation model for a relationship between mud density, temperature and mud rheology.
 19. The system of claim 11, in which the historical well event comprises at least one of the group consisting of wellbore collapse, wellbore instability and stuck pipe.
 20. The system of claim 11, in which the predictive model is configured to predict horizontal stresses in an earth formation of the target well.
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